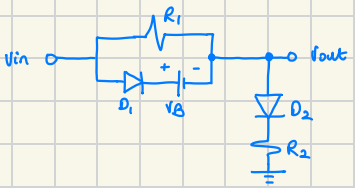
ELG 2136

Lecture 3

Graphical analysis of circuit with diodes + analytical intuition.

a) $v_{out} = ?$ if $v_{in} = 6V$



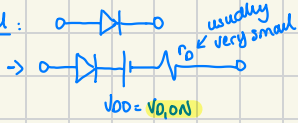
$V_{D,ON} = 0.7\,[V]$

$V_B = 3\,[V]$

$R_1 = 6000\,[\Omega]$

$R_2 = 2000\,[\Omega]$

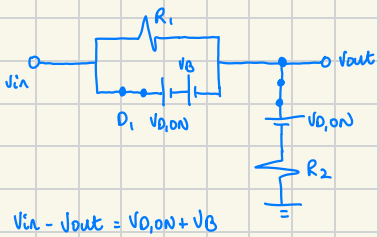
Real diode model:



-> r_D can be ignored.

Assumption: D_1 & D_2 both on.

-> circuit becomes



$v_{in} - v_{out} = V_{D,ON} + V_B$

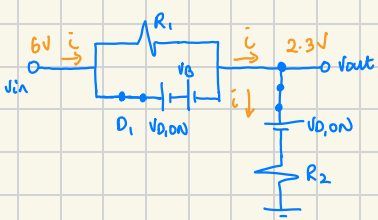
-> $v_{in} = V_{D,ON} + V_B + v_{out}$

-> $\boxed{v_{out} = v_{in} - V_{D,ON} - V_B}$ -> $v_{out} = 6 - 0.7 - 3 = 2.3\,[V]$

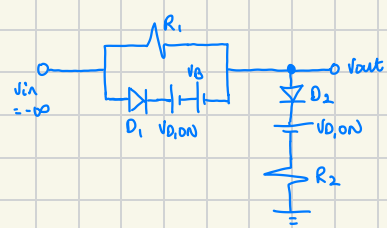
-> $\boxed{v_{out} = 2.3\,[V]}$

Confirm the assumption [making sure there are no contradictions]

$v_{in} > v_{out}$



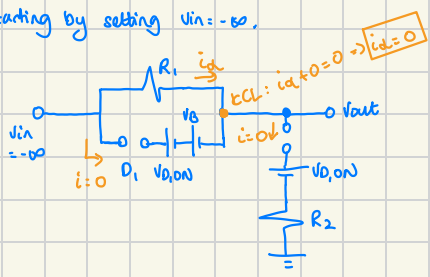
b) Plot voltage transfer characteristics (v_{out} vs v_{in}), starting by setting $v_{in} = -\infty$.



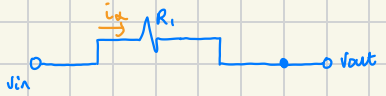
D_1: OFF

D_2: OFF

->

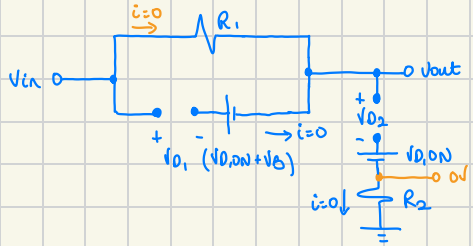


Cleaning up:



$V_{out} = V_{in}$

Finding breakpoints D_1: OFF & D_2: OFF, at what point will D_1 and/or D_2 switch modes.



$V_{out} = V_{in}$

$V_{in} - V_{out} = V_{D_1} + V_{D,ON} + V_B$

$\Rightarrow V_{D_1} = V_{in} - V_{out} - V_{D,ON} - V_B$

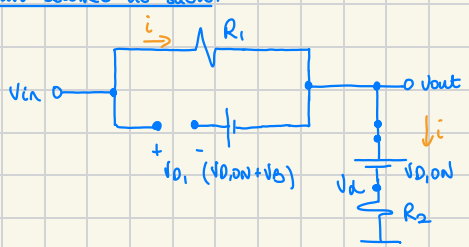
$= \cancel{V_{in}} - \cancel{V_{in}} - V_{D,ON} - V_B = -(V_{D,ON} + V_B)$ [no breakpoint] constant voltage.

$V_{D_2} = V_{out} - V_{D,ON}$, $V_{out} = V_{in}$

$\Rightarrow \boxed{V_{D_2} = V_{in} - V_{D,ON}}$ Condition for D_2 to be on: $V_{D_2} = 0$

$\Rightarrow V_{in} - V_{D,ON} = 0 \Rightarrow \boxed{V_{in} = V_{D,ON}} \Rightarrow \boxed{V_{in} = 0.7\,[V]}$

Circuit becomes as such:



$i = \frac{V_{in} - V_{out}}{R_1}$, $i = \frac{V_d - 0}{R_2} = \frac{V_d}{R_2}$, $V_{D,ON} = V_{out} - V_d$

$\Rightarrow V_{D,ON} + V_d = V_{out}$

$\Rightarrow \boxed{V_d = V_{out} - V_{D,ON}}$

$i = \frac{V_{out} - V_{D,ON}}{R_2}$

$\Rightarrow \frac{V_{in} - V_{out}}{R_1} = \frac{V_{out} - V_{D,ON}}{R_2} \Rightarrow \frac{V_{in}}{R_1} - \frac{V_{out}}{R_1} = \frac{V_{out}}{R_2} - \frac{V_{D,ON}}{R_2}$

$\Rightarrow \frac{V_{in}}{R_1} + \frac{V_{D,ON}}{R_2} = \frac{V_{out}}{R_2} + \frac{V_{out}}{R_1} \Rightarrow \frac{V_{in}}{R_1} + \frac{V_{D,ON}}{R_2} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) V_{out} = \left(\frac{R_2 + R_1}{R_1 R_2}\right) V_{out}$

$\Rightarrow V_{out} = \left(\frac{R_1 R_2}{R_1 + R_2}\right)\left(\frac{V_{in}}{R_1} + \frac{V_{D,ON}}{R_2}\right) = \left(\frac{\cancel{R_1} R_2}{R_1 + R_2}\right)\left(\frac{1}{\cancel{R_1}}\right) V_{in} + \left(\frac{R_1 \cancel{R_2}}{R_1 + R_2}\right)\left(\frac{1}{\cancel{R_2}}\right) V_{D,ON}$

$\Rightarrow \boxed{V_{out} = \frac{R_2}{R_1 + R_2} V_{in} + \frac{R_1}{R_1 + R_2} V_{D,ON}} \Rightarrow V_{out} = \left(\frac{2000}{6000 + 2000}\right) V_{in} + \left(\frac{6000}{6000 + 2000}\right) V_{D,ON}$

$\Rightarrow \boxed{V_{out} = \frac{1}{4} V_{in} + \frac{21}{40}}$

Finding Breakpoints

$V_{in} - V_{out} = V_{D_1} + V_{D,ON} + V_B$

$\Rightarrow V_{D_1} = V_{in} - V_{out} - V_{D,ON} - V_B$

$= V_{in} - \left[\left(\frac{R_2}{R_1+R_2}\right)V_{in} + \left(\frac{R_1}{R_1+R_2}\right)V_{D,ON}\right] - V_{D,ON} - V_B = V_{in} - \frac{R_2}{R_1+R_2}V_{in} - \frac{R_1}{R_1+R_2}V_{D,ON} - V_{D,ON} - V_B$

$= \left(1 - \frac{R_2}{R_1+R_2}\right)V_{in} - \left(\frac{R_1}{R_1+R_2} + 1\right)V_{D,ON} - V_B = \frac{R_1+R_2-R_2}{R_1+R_2}V_{in} - \left(\frac{R_1+R_1+R_2}{R_1+R_2}\right)V_{D,ON} - V_B$

$= \frac{R_1}{R_1+R_2}V_{in} - \left(\frac{2R_1+R_2}{R_1+R_2}V_{D,ON} + V_B\right)$

$\Rightarrow \boxed{V_{D_1} = \frac{R_1}{R_1+R_2}V_{in} - \left(\frac{2R_1+R_2}{R_1+R_2}V_{D,ON} + V_B\right)}$ Condition for D_1 to be ON: $V_{D_1} = 0$

$\Rightarrow \frac{R_1}{R_1+R_2}V_{in} - \left(\frac{2R_1+R_2}{R_1+R_2}V_{D,ON} + V_B\right) = 0$

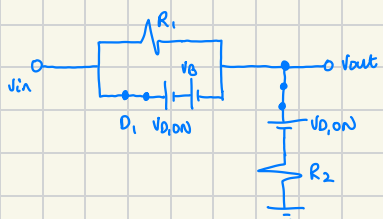
$\Rightarrow \frac{R_1}{R_1+R_2}V_{in} = \left(\frac{2R_1+R_2}{R_1+R_2}V_{D,ON} + V_B\right) \Rightarrow V_{in} = \frac{(R_1+R_2)}{R_1}\frac{(2R_1+R_2)}{(R_1+R_2)}V_{D,ON} + \frac{R_1+R_2}{R_1}V_B \Rightarrow \boxed{V_{in} = \frac{2R_1+R_2}{R_1}V_{D,ON} + \frac{(R_1+R_2)}{R_1}V_B}$

Breakpoint: Here, D_1: OFF → ON

$\Rightarrow V_{in} = \left[\frac{2(6000)+2000}{6000}\right](0.7) + \left(\frac{6000+2000}{6000}\right)(3) = \frac{169}{30} = 5.63$

$\Rightarrow \boxed{V_{in} = \frac{169}{30}[V] = 5.63[V]}$

Circuit becomes as such:

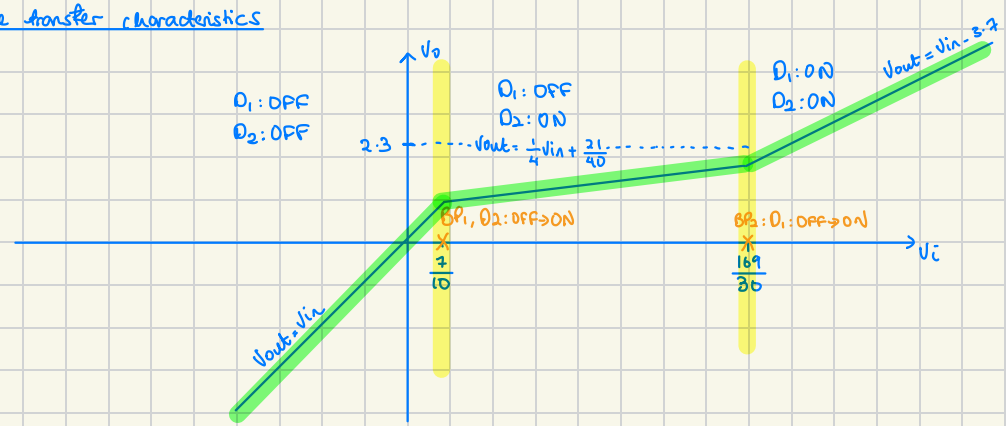


same as before: $\boxed{V_{out} = V_{in} - V_{D,ON} - V_B}$

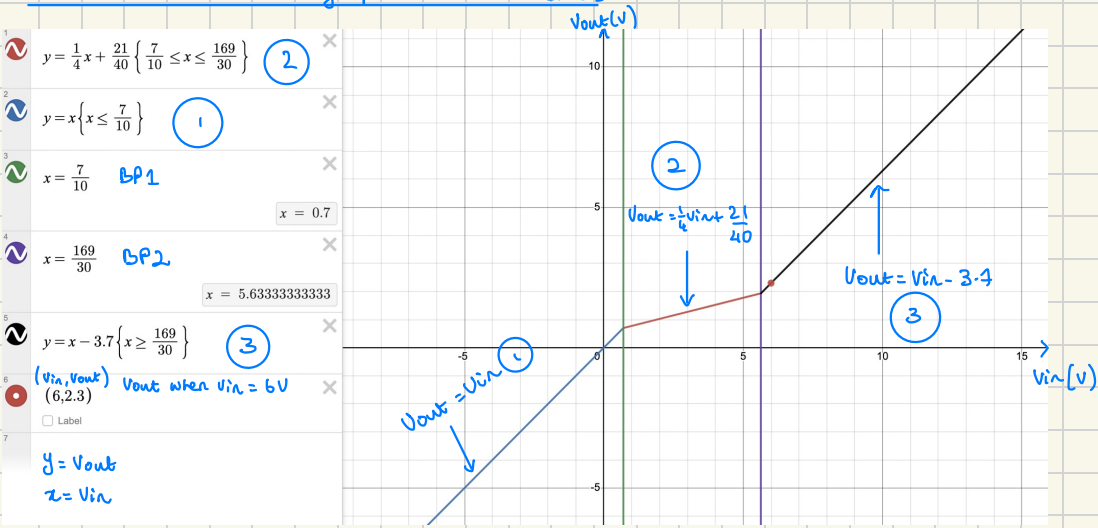
$\Rightarrow V_{out} = V_{in} - 0.7 - 3$

$\Rightarrow \boxed{V_{out} = V_{in} - 3.7}$

Voltage transfer characteristics



Below is a cleaner graph from Desmos:



BP 1

@ $V_{in} = \frac{7}{10} V$

D_2: OFF → ON

BP2

@ $V_{in} = \frac{169}{30} V$

D_1: OFF → ON